

■ 2.1.14 Advanced Topic: Making Lists out of Functions

<code>Array[f, n]</code>	generate a length n list of the form $\{f[1], f[2], \dots\}$
<code>Array[f, {$n_1, n_2, \dots$}]</code>	generate a $n_1 \times n_2 \times \dots$ nested list, each of whose entries consists of f applied to its indices
<code>NestList[f, x, n]</code>	generate a list of the form $\{x, f[x], f[f[x]], \dots\}$, where f is nested up to n deep
<code>Accumulate[f, list]</code>	cumulatively applies f to pairs of elements in $list$

Making lists from functions.

This makes a list of 5 elements, each of the form $p[i]$.

```
In[1]:= Array[p, 5]
Out[1]= {p[1], p[2], p[3], p[4], p[5]}
```

Here is another way to produce the same list.

```
In[2]:= Table[p[i], {i, 5}]
Out[2]= {p[1], p[2], p[3], p[4], p[5]}
```

This produces a list whose elements are $i + i^2$.

```
In[3]:= Array[Function[i, i + i^2], 5]
Out[3]= {2, 6, 12, 20, 30}
```

This generates a 2×3 matrix whose entries are $m[i, j]$.

```
In[4]:= Array[m, {2, 3}]
Out[4]= {{m[1, 1], m[1, 2], m[1, 3]},
         {m[2, 1], m[2, 2], m[2, 3]}}
```

This produces a list of the first four “iterates” of f . The list includes the argument x itself, so has length 5.

```
In[5]:= NestList[f, x, 4]
Out[5]= {x, f[x], f[f[x]], f[f[f[x]]], f[f[f[f[x]]]]}
```

This gives a list of the first three iterated derivatives of x^n .

```
In[6]:= NestList[Function[y, D[y, x]], x^n, 3]
Out[6]= {x^n, n x^{n-1}, n(n-1) x^{n-2}, n(n-1)(n-2) x^{n-3}}
```

Functions like `NestList` and `Map` allow you to apply functions to single expressions. `Array` and `Apply` can deal with sequences of any number of expressions. `Accumulate` is intended specifically for dealing with functions that take exactly two arguments.

Here is a list of four elements.

```
In[7]:= Range[4]  
Out[7]= {1, 2, 3, 4}
```

This shows the result of “accumulating” `f` over the list.

```
In[8]:= Accumulate[f, %]  
Out[8]= {1, f[1, 2], f[f[1, 2], 3], f[f[f[1, 2], 3], 4]}
```

If you accumulate `Plus` over a list, you get a list of cumulative sums.

```
In[9]:= Accumulate[Plus, Range[5]]  
Out[9]= {1, 3, 6, 10, 15}
```